

Commutative Algebra 1

1 Tensoring with A/I : detailed explanation of the four examples

The common principle behind all four examples is the canonical isomorphism

$$(A/I) \otimes_A M \cong M/IM,$$

where:

- A is a ring,
- $I \subseteq A$ is an ideal,
- M is an A -module,
- IM is the submodule generated by all products im with $i \in I$ and $m \in M$.

Intuitively, tensoring with A/I means that every element of the ideal I is forced to act as 0. In other words, we are “reducing the module M modulo I .”

1.1 Element-level meaning of $(A/I) \otimes_A M$

An element of $(A/I) \otimes_A M$ is a finite sum of pure tensors

$$(\bar{a}) \otimes m,$$

where $\bar{a} = a + I \in A/I$ and $m \in M$.

Because $\bar{a} = a \cdot \bar{1}$ in A/I , we have

$$(\bar{a}) \otimes m = (\bar{1}) \otimes am.$$

So every pure tensor can be rewritten in the form

$$\bar{1} \otimes m.$$

Now if $i \in I$, then $\bar{i} = 0$ in A/I , hence

$$\bar{1} \otimes (im) = \bar{i} \otimes m = 0.$$

Therefore every element of IM becomes zero in the tensor product.

This is exactly why the tensor product depends only on the class of m modulo IM . The isomorphism is:

$$\Phi : (A/I) \otimes_A M \longrightarrow M/IM, \quad (\bar{a}) \otimes m \longmapsto am + IM.$$

Its inverse is given by

$$\Psi : M/IM \longrightarrow (A/I) \otimes_A M, \quad m + IM \longmapsto \bar{1} \otimes m.$$

So the tensor product remembers precisely the part of M that survives after we impose the rule $I = 0$.

1.2 What survives and what dies

Under the passage from M to $(A/I) \otimes_A M$:

- terms not involving the ideal I survive,
- all terms in the submodule IM vanish.

Thus:

- over $\mathbb{Z}$, tensoring with $\mathbb{Z}/m$ means reducing modulo m ,
- over $k[x]$, tensoring with $k[x]/(x)$ means setting $x = 0$.

2 Example 1: $A = \mathbb{Z}$, $I = (m)$, $M = \mathbb{Z}^r$

We compute

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} \mathbb{Z}^r.$$

By the general formula,

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} \mathbb{Z}^r \cong \mathbb{Z}^r / m\mathbb{Z}^r.$$

Now

$$m\mathbb{Z}^r = \{(ma_1, \dots, ma_r) : a_1, \dots, a_r \in \mathbb{Z}\}.$$

So $\mathbb{Z}^r / m\mathbb{Z}^r$ is obtained by reducing each coordinate modulo m . Therefore

$$\mathbb{Z}^r / m\mathbb{Z}^r \cong (\mathbb{Z}/m)^r.$$

Hence

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} \mathbb{Z}^r \cong (\mathbb{Z}/m)^r.$$

2.1 Element-wise description

Let $e_1, \dots, e_r$ be the standard basis of $\mathbb{Z}^r$. Any vector $v \in \mathbb{Z}^r$ can be written as

$$v = a_1 e_1 + \dots + a_r e_r.$$

Then

$$[1]_m \otimes v = [1]_m \otimes (a_1 e_1 + \dots + a_r e_r) = [a_1]_m \otimes e_1 + \dots + [a_r]_m \otimes e_r.$$

So the tensor product simply remembers the coefficients modulo m .

2.2 Concrete example

Take $m = 5$, $r = 2$, and $v = (7, 11)$. Then

$$(7, 11) \equiv (2, 1) \pmod{5}.$$

Thus

$$[1]_5 \otimes (7, 11) \text{ corresponds to } ([2]_5, [1]_5) \in (\mathbb{Z}/5)^2.$$

2.3 Interpretation

The “vectors” here are genuine integer vectors in $\mathbb{Z}^r$, but after tensoring with $\mathbb{Z}/m$, only their coordinates modulo m remain visible.

3 Example 2: $A = \mathbb{Z}$, $I = (m)$, $M = \mathbb{Z}/n$

We compute

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} (\mathbb{Z}/n).$$

Using the same principle,

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} (\mathbb{Z}/n) \cong (\mathbb{Z}/n)/m(\mathbb{Z}/n).$$

So we must understand the subgroup $m(\mathbb{Z}/n)$.

3.1 Computing $m(\mathbb{Z}/n)$

Inside $\mathbb{Z}/n$, multiplication by m gives the subgroup

$$m(\mathbb{Z}/n) = \{[mk]_n : k \in \mathbb{Z}\}.$$

This is the cyclic subgroup generated by $[m]_n$.

The order of $[m]_n$ in $\mathbb{Z}/n$ is

$$\frac{n}{\gcd(m, n)},$$

so the subgroup $m(\mathbb{Z}/n)$ has exactly that many elements.

Therefore the quotient $(\mathbb{Z}/n)/m(\mathbb{Z}/n)$ has cardinality

$$\frac{n}{n/\gcd(m, n)} = \gcd(m, n).$$

Since the quotient of a cyclic group is cyclic, we get

$$(\mathbb{Z}/n)/m(\mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n).$$

Hence

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} (\mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n).$$

3.2 Element-wise description

The tensor product is generated by the single element

$$[1]_m \otimes [1]_n.$$

Indeed, for any $a, b \in \mathbb{Z}$,

$$[a]_m \otimes [b]_n = ab([1]_m \otimes [1]_n).$$

So the whole tensor product is cyclic.

Moreover,

$$m([1]_m \otimes [1]_n) = 0 \quad \text{and} \quad n([1]_m \otimes [1]_n) = 0,$$

so the order of the generator divides both m and n , hence divides $\gcd(m, n)$. In fact, its exact order is $\gcd(m, n)$.

3.3 Concrete example

Let $m = 12$ and $n = 18$. Then

$$\gcd(12, 18) = 6,$$

so

$$(\mathbb{Z}/12) \otimes_{\mathbb{Z}} (\mathbb{Z}/18) \cong \mathbb{Z}/6.$$

The element

$$[1]_{12} \otimes [1]_{18}$$

is a generator of this cyclic group.

3.4 Interpretation

This tensor product retains only the *common torsion* visible in both cyclic modules:

- $\mathbb{Z}/m$ has m -torsion,
- $\mathbb{Z}/n$ has n -torsion,
- their tensor product detects the part compatible with both, namely $\gcd(m, n)$ -torsion.

4 Example 3: $A = k[x]$, $I = (x)$ — evaluation at $x = 0$

Now let

$$A = k[x], \quad I = (x).$$

Then

$$A/I = k[x]/(x).$$

But quotienting by (x) kills all multiples of x , so every polynomial becomes equal to its constant term. Thus

$$k[x]/(x) \cong k,$$

via the map

$$f(x) \mapsto f(0).$$

So in this situation,

$$A/I \cong k.$$

Therefore for any A -module M ,

$$k \otimes_{k[x]} M \cong M/xM.$$

4.1 Why this means “set $x = 0$ ”

Since $x = 0$ in the quotient ring $k[x]/(x)$, we have in the tensor product:

$$1 \otimes (xm) = x \otimes m = 0.$$

So every element of the form xm vanishes. That is why we divide by xM .

This is exactly the module-theoretic meaning of evaluating at $x = 0$: all x -terms disappear.

4.2 Particular case: $M = k[x]$

If $M = A = k[x]$, then

$$k \otimes_{k[x]} k[x] \cong k[x]/xk[x].$$

But $xk[x]$ is the set of polynomials with zero constant term, so the quotient keeps only the constant term:

$$k[x]/xk[x] \cong k.$$

Thus

$$k \otimes_{k[x]} k[x] \cong k.$$

4.3 Element-wise example

Take the polynomial

$$f(x) = 3 + 2x + x^2.$$

Then in the tensor product,

$$1 \otimes f(x) = 1 \otimes (3 + 2x + x^2).$$

Since all terms involving x vanish,

$$1 \otimes (3 + 2x + x^2) = 1 \otimes 3.$$

So only the constant term remains.

4.4 Interpretation

Tensoring with $k \cong k[x]/(x)$ means:

- forget all multiples of x ,
- keep only what survives at $x = 0$,
- algebraically: pass from M to the quotient M/xM .

5 Example 4: $A = k[x]$, $I = (x)$, $M = A/(x^2)$

Now let

$$A = k[x], \quad I = (x), \quad M = A/(x^2) = k[x]/(x^2).$$

We want to compute

$$(A/I) \otimes_A M = k \otimes_{k[x]} A/(x^2).$$

By the general rule,

$$(A/I) \otimes_A M \cong M/xM.$$

So we must compute xM .

5.1 Structure of $M = A/(x^2)$

In the quotient $k[x]/(x^2)$, every polynomial is equivalent to one of the form

$$a + bx \pmod{x^2}.$$

So as a k -vector space,

$$M = k \cdot 1 \oplus k \cdot x.$$

A convenient basis is

$$\{1, x\}.$$

However, inside this module we have

$$x^2 = 0.$$

So x is not zero, but it is nilpotent.

5.2 Computing xM

Take an arbitrary element

$$a + bx \in M.$$

Multiplying by x , we get

$$x(a + bx) = ax + bx^2.$$

Since $x^2 = 0$ in M , this becomes

$$x(a + bx) = ax.$$

Therefore

$$xM = kx.$$

So the quotient is

$$M/xM = (k \cdot 1 \oplus k \cdot x)/(k \cdot x) \cong k.$$

Hence

$$k \otimes_{k[x]} A/(x^2) \cong M/xM \cong k.$$

5.3 Element-wise tensor computation

Take the class of $a + bx$ in $M = A/(x^2)$. Then in the tensor product,

$$1 \otimes (a + bx) = 1 \otimes a + 1 \otimes bx.$$

But

$$1 \otimes bx = b(1 \otimes x).$$

Now $x = 0$ in $k = A/(x)$, so

$$1 \otimes x = x \otimes 1 = 0.$$

Therefore

$$1 \otimes (a + bx) = 1 \otimes a.$$

So again, only the constant part survives.

5.4 Interpretation

This example is important because $M = A/(x^2)$ is larger than k : it has an extra “infinitesimal” direction kx . But after tensoring with $k = A/(x)$, the x -part disappears.

So even though $x \neq 0$ in M , it becomes invisible after imposing $x = 0$.

6 What are the “vectors” or “elements” in these examples?

These are not always vectors in the geometric sense. They are simply *elements of modules*.

- In Example 1, the elements are tuples

$$(a_1, \dots, a_r) \in \mathbb{Z}^r.$$

These behave like integer vectors.

- In Example 2, the elements are residue classes

$$[a]_n \in \mathbb{Z}/n.$$

- In Example 3, the elements are arbitrary module elements $m \in M$, where M is any $k[x]$ -module.
- In Example 4, the elements are classes

$$a + bx \pmod{x^2},$$

so each element has a constant part and an x -part.

In all cases, after tensoring with A/I , every element can be represented by something of the form

$$\bar{1} \otimes m,$$

and two elements become equal precisely when they differ by something from IM .

Thus the tensor product is measuring the module M after “forgetting” everything killed by the ideal I .

7 One-line intuition for each example

- **Example 1:** reduce each coordinate modulo m .
- **Example 2:** keep only the common torsion shared by $\mathbb{Z}/m$ and $\mathbb{Z}/n$.
- **Example 3:** set $x = 0$ in the module.
- **Example 4:** set $x = 0$ even in a module where $x \neq 0$ but $x^2 = 0$, so the nilpotent x -part disappears.

8 Final universal picture

The construction

$$(A/I) \otimes_A M$$

means that we are viewing the module M after passing from the ring A to the quotient ring A/I .

Equivalently:

- coefficients are reduced modulo I ,
- every element of I acts as zero,
- all elements of IM vanish,
- the remaining information is exactly the quotient module M/IM .

So the identity

$$(A/I) \otimes_A M \cong M/IM$$

is the precise algebraic statement that tensoring with A/I is the same as reducing the module M modulo the action of the ideal I .

Problem 5 — Zero divisors, localization, and associated primes via $D(A)$

Exact statement. In a ring A , let $D(A)$ denote the set of prime ideals $\mathfrak{p}$ which satisfy the following condition: there exists an $a \in A$ such that $\mathfrak{p}$ is minimal in $V(\text{Ann}(a))$.

- (a) Show that x is a zero divisor in A if and only if $x \in \mathfrak{p}$ for some $\mathfrak{p} \in D(A)$.
- (b) Let S be a multiplicatively closed subset of A . Under the canonical identification of $\text{Spec } S^{-1}A$ with a subset of $\text{Spec } A$, show that

$$D(S^{-1}A) = D(A) \cap \text{Spec } S^{-1}A.$$

- (c) If the 0 ideal has a primary decomposition, show that $D(A)$ is the set of associated primes of A .

Solution.

- (a) First suppose that x is a zero divisor. Then there exists $a \neq 0$ such that

$$xa = 0.$$

Hence $x \in \text{Ann}(a)$. Choose a prime ideal $\mathfrak{p}$ minimal over $\text{Ann}(a)$. Then $\mathfrak{p} \in D(A)$ by definition, and $\text{Ann}(a) \subseteq \mathfrak{p}$, so $x \in \mathfrak{p}$.

Conversely, suppose $x \in \mathfrak{p}$ for some $\mathfrak{p} \in D(A)$. Then there exists $a \in A$ such that $\mathfrak{p}$ is minimal over $\text{Ann}(a)$. Let

$$I = \text{Ann}(a), \quad B = A/I.$$

Then $\mathfrak{q} := \mathfrak{p}/I$ is a minimal prime of B , and $\bar{x} \in \mathfrak{q}$.

Localizing at $\mathfrak{q}$, the ring $B_{\mathfrak{q}}$ has only one prime ideal, namely $\mathfrak{q}B_{\mathfrak{q}}$. Hence every element of $\mathfrak{q}B_{\mathfrak{q}}$ is nilpotent. In particular, $\bar{x}/1$ is nilpotent in $B_{\mathfrak{q}}$. Thus there exist $s \notin \mathfrak{q}$ and $n \geq 1$ such that

$$s\bar{x}^n = 0 \quad \text{in } B.$$

Choose n minimal with this property. Then

$$\bar{b} := s\bar{x}^{n-1} \neq 0, \quad \bar{x}\bar{b} = 0.$$

So $\bar{x}$ is a zero divisor in B . Lift $\bar{b}$ to some $b \in A$. Since $\bar{b} \neq 0$, we have $b \notin I = \text{Ann}(a)$, so $ba \neq 0$. Also $\bar{x}\bar{b} = 0$ means $xb \in I$, hence

$$x(ba) = (xb)a = 0.$$

Therefore x is a zero divisor in A .

Thus

$$x \text{ is a zero divisor in } A \iff x \in \mathfrak{p} \text{ for some } \mathfrak{p} \in D(A).$$

(b) Identify $\text{Spec } S^{-1}A$ with the set of prime ideals $\mathfrak{p} \subset A$ such that $\mathfrak{p} \cap S = \emptyset$.

First let $\mathfrak{q} \in D(S^{-1}A)$. Then there exists $a/s \in S^{-1}A$ such that $\mathfrak{q}$ is minimal over

$$\text{Ann}_{S^{-1}A}(a/s) = S^{-1} \text{Ann}_A(a).$$

Write $\mathfrak{q} = S^{-1}\mathfrak{p}$, where $\mathfrak{p} \cap S = \emptyset$. We claim that $\mathfrak{p}$ is minimal over $\text{Ann}_A(a)$. Indeed, if $\mathfrak{r} \subsetneq \mathfrak{p}$ is a prime ideal containing $\text{Ann}_A(a)$, then $\mathfrak{r} \cap S = \emptyset$ because $\mathfrak{r} \subseteq \mathfrak{p}$. Hence $S^{-1}\mathfrak{r}$ is a prime of $S^{-1}A$ properly contained in $\mathfrak{q}$ and containing $S^{-1} \text{Ann}_A(a)$, contradiction. So $\mathfrak{p} \in D(A)$, and $\mathfrak{p} \in \text{Spec } S^{-1}A$. Therefore

$$D(S^{-1}A) \subseteq D(A) \cap \text{Spec } S^{-1}A.$$

Conversely, let $\mathfrak{p} \in D(A) \cap \text{Spec } S^{-1}A$. Then $\mathfrak{p} \cap S = \emptyset$, and there exists $a \in A$ such that $\mathfrak{p}$ is minimal over $\text{Ann}_A(a)$. Since $\text{Ann}_A(a) \subseteq \mathfrak{p}$ and $\mathfrak{p} \cap S = \emptyset$, the element $a/1$ is nonzero in $S^{-1}A$. Also

$$\text{Ann}_{S^{-1}A}(a/1) = S^{-1} \text{Ann}_A(a).$$

If $S^{-1}\mathfrak{r} \subsetneq S^{-1}\mathfrak{p}$ is a prime containing $S^{-1} \text{Ann}_A(a)$, then $\mathfrak{r} \subsetneq \mathfrak{p}$ is a prime containing $\text{Ann}_A(a)$, contradiction. Hence $S^{-1}\mathfrak{p} \in D(S^{-1}A)$. Thus

$$D(A) \cap \text{Spec } S^{-1}A \subseteq D(S^{-1}A).$$

Therefore

$$D(S^{-1}A) = D(A) \cap \text{Spec } S^{-1}A.$$

(c) Assume that

$$0 = Q_1 \cap \cdots \cap Q_n$$

is an irredundant primary decomposition, where each Q_i is $\mathfrak{p}_i$ -primary. Then

$$\text{Ass}_A(A) = \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}.$$

We show that $D(A)$ is the same set.

First, let $\mathfrak{p}_i$ be one of these primes. Since the decomposition is irredundant, there exists

$$a_i \in \bigcap_{j \neq i} Q_j \setminus Q_i.$$

Then

$$\text{Ann}(a_i) = \{r \in A : ra_i = 0\}.$$

Because $a_i \in Q_j$ for all $j \neq i$, the condition $ra_i = 0$ is equivalent to $ra_i \in Q_i$. Hence

$$\text{Ann}(a_i) = (Q_i : a_i).$$

Since Q_i is $\mathfrak{p}_i$ -primary and $a_i \notin Q_i$, the colon ideal $(Q_i : a_i)$ is $\mathfrak{p}_i$ -primary. Therefore

$$\sqrt{\text{Ann}(a_i)} = \mathfrak{p}_i,$$

so $\mathfrak{p}_i$ is the unique minimal prime over $\text{Ann}(a_i)$. Hence $\mathfrak{p}_i \in D(A)$.

Thus

$$\text{Ass}_A(A) \subseteq D(A).$$

Conversely, let $\mathfrak{p} \in D(A)$. Then $\mathfrak{p}$ is minimal over $\text{Ann}(a)$ for some $a \in A$. Since

$$0 = \bigcap_{i=1}^n Q_i,$$

we obtain

$$\text{Ann}(a) = \bigcap_{i=1}^n (Q_i : a).$$

For each i , either $a \in Q_i$, in which case $(Q_i : a) = A$, or $a \notin Q_i$, in which case $(Q_i : a)$ is $\mathfrak{p}_i$ -primary. Therefore $\text{Ann}(a)$ has a primary decomposition whose associated primes are among the $\mathfrak{p}_i$. Hence every minimal prime over $\text{Ann}(a)$, in particular $\mathfrak{p}$, must equal one of the $\mathfrak{p}_i$. So

$$D(A) \subseteq \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\} = \text{Ass}_A(A).$$

Therefore

$$D(A) = \text{Ass}_A(A).$$

Problem 6 — The ideals $S_{\mathfrak{p}}(0)$ and minimal primes

Exact statement. For $\mathfrak{p} \subset A$ a prime ideal, denote by $S_{\mathfrak{p}}(0)$ the kernel of the canonical homomorphism

$$A \rightarrow A_{\mathfrak{p}}.$$

Prove:

- (a) $S_{\mathfrak{p}}(0) \subseteq \mathfrak{p}$.
- (b) If $\mathfrak{p} \supseteq \mathfrak{p}'$, then $S_{\mathfrak{p}}(0) \subseteq S_{\mathfrak{p}'}(0)$.
- (c) $\sqrt{S_{\mathfrak{p}}(0)} = \mathfrak{p}$ if and only if $\mathfrak{p}$ is a minimal prime ideal of A .

- (d) $\bigcap_{\mathfrak{p} \in D(A)} S_{\mathfrak{p}}(0) = 0$, where $D(A)$ is the set defined in the previous exercise.

Solution.

Recall that

$$S_{\mathfrak{p}}(0) = \{a \in A : \exists s \notin \mathfrak{p} \text{ such that } sa = 0\}.$$

(a) Let $a \in S_{\mathfrak{p}}(0)$. Then $sa = 0$ for some $s \notin \mathfrak{p}$. Since $0 \in \mathfrak{p}$, we have $sa \in \mathfrak{p}$. Because $\mathfrak{p}$ is prime and $s \notin \mathfrak{p}$, it follows that $a \in \mathfrak{p}$. Hence

$$S_{\mathfrak{p}}(0) \subseteq \mathfrak{p}.$$

(b) Suppose $\mathfrak{p} \supseteq \mathfrak{p}'$, and let $a \in S_{\mathfrak{p}}(0)$. Then there exists $s \notin \mathfrak{p}$ such that $sa = 0$. Since $\mathfrak{p}' \subseteq \mathfrak{p}$, we also have $s \notin \mathfrak{p}'$. Thus $a \in S_{\mathfrak{p}'}(0)$. Therefore

$$S_{\mathfrak{p}}(0) \subseteq S_{\mathfrak{p}'}(0).$$

(c) First assume that $\mathfrak{p}$ is a minimal prime ideal of A . By part (a),

$$S_{\mathfrak{p}}(0) \subseteq \mathfrak{p},$$

so

$$\sqrt{S_{\mathfrak{p}}(0)} \subseteq \mathfrak{p}.$$

Now let $x \in \mathfrak{p}$. In the localized ring $A_{\mathfrak{p}}$, the prime ideals correspond to prime ideals of A contained in $\mathfrak{p}$. Since $\mathfrak{p}$ is minimal, the only prime ideal of $A_{\mathfrak{p}}$ is $\mathfrak{p}A_{\mathfrak{p}}$. Therefore the nilradical of $A_{\mathfrak{p}}$ is $\mathfrak{p}A_{\mathfrak{p}}$, so $x/1$ is nilpotent in $A_{\mathfrak{p}}$. Hence for some $n \geq 1$,

$$(x/1)^n = 0,$$

which means $x^n \in S_{\mathfrak{p}}(0)$. Thus $x \in \sqrt{S_{\mathfrak{p}}(0)}$, and so

$$\mathfrak{p} \subseteq \sqrt{S_{\mathfrak{p}}(0)}.$$

Therefore

$$\sqrt{S_{\mathfrak{p}}(0)} = \mathfrak{p}.$$

Conversely, suppose

$$\sqrt{S_{\mathfrak{p}}(0)} = \mathfrak{p}.$$

If $\mathfrak{p}$ were not minimal, there would exist a prime ideal $\mathfrak{q} \subsetneq \mathfrak{p}$. We claim that

$$S_{\mathfrak{p}}(0) \subseteq \mathfrak{q}.$$

Indeed, if $a \in S_{\mathfrak{p}}(0)$, then $sa = 0$ for some $s \notin \mathfrak{p}$. Since $\mathfrak{q} \subseteq \mathfrak{p}$, also $s \notin \mathfrak{q}$. Because $\mathfrak{q}$ is prime and $sa = 0 \in \mathfrak{q}$, it follows that $a \in \mathfrak{q}$. Therefore

$$S_{\mathfrak{p}}(0) \subseteq \mathfrak{q},$$

hence

$$\sqrt{S_{\mathfrak{p}}(0)} \subseteq \mathfrak{q} \subsetneq \mathfrak{p},$$

contradiction. Thus $\mathfrak{p}$ must be minimal.

So

$$\sqrt{S_{\mathfrak{p}}(0)} = \mathfrak{p} \iff \mathfrak{p} \text{ is a minimal prime ideal of } A.$$

(d) We prove that

$$\bigcap_{\mathfrak{p} \in D(A)} S_{\mathfrak{p}}(0) = 0.$$

Let $x \neq 0$. Choose a prime ideal $\mathfrak{p}$ minimal over $\text{Ann}(x)$. Then $\mathfrak{p} \in D(A)$ by definition. We claim that $x \notin S_{\mathfrak{p}}(0)$. Otherwise there would exist $s \notin \mathfrak{p}$ such that

$$sx = 0.$$

Then $s \in \text{Ann}(x) \subseteq \mathfrak{p}$, contradiction. Hence $x \notin S_{\mathfrak{p}}(0)$.

Therefore no nonzero element belongs to every $S_{\mathfrak{p}}(0)$, and we conclude that

$$\bigcap_{\mathfrak{p} \in D(A)} S_{\mathfrak{p}}(0) = 0.$$

Problem 7 — A local-global criterion for Noetherianity

Exact statement. Let A be a ring such that

- (1) for every maximal ideal $\mathfrak{m}$ of A , the local ring $A_{\mathfrak{m}}$ is Noetherian;
- (2) for each non-zero $x \in A$, the set of maximal ideals of A which contain x is finite.

Show that A is Noetherian.

Solution.

We show that every ideal of A is finitely generated.

Let $I \subseteq A$ be an ideal. If $I = 0$, there is nothing to prove. So assume $I \neq 0$, and choose $0 \neq x \in I$.

By assumption (2), only finitely many maximal ideals contain x . Denote them by

$$\mathfrak{m}_1, \dots, \mathfrak{m}_r.$$

For each i , the ideal $I_{\mathfrak{m}_i}$ of the Noetherian ring $A_{\mathfrak{m}_i}$ is finitely generated. Hence there exist elements

$$a_{i1}, \dots, a_{in_i} \in I$$

such that

$$I_{\mathfrak{m}_i} = (a_{i1}, \dots, a_{in_i})_{\mathfrak{m}_i}.$$

Now define

$$J = (x, a_{11}, \dots, a_{1n_1}, \dots, a_{r1}, \dots, a_{rn_r}) \subseteq I.$$

This is a finitely generated ideal. We claim that $I = J$.

Let $\mathfrak{m}$ be any maximal ideal of A .

If $x \notin \mathfrak{m}$, then $x/1$ is a unit in $A_{\mathfrak{m}}$. Since $x \in J \subseteq I$, both $J_{\mathfrak{m}}$ and $I_{\mathfrak{m}}$ contain a unit, hence

$$J_{\mathfrak{m}} = A_{\mathfrak{m}} = I_{\mathfrak{m}}.$$

If $x \in \mathfrak{m}$, then $\mathfrak{m} = \mathfrak{m}_i$ for some i , and by construction

$$J_{\mathfrak{m}} = I_{\mathfrak{m}}.$$

Thus

$$J_{\mathfrak{m}} = I_{\mathfrak{m}} \quad \text{for every maximal ideal } \mathfrak{m}.$$

Consider the quotient module I/J . For every maximal ideal $\mathfrak{m}$,

$$(I/J)_{\mathfrak{m}} = 0.$$

A module is zero if and only if all its localizations at maximal ideals are zero. Therefore $I/J = 0$, so $I = J$.

Hence every ideal of A is finitely generated. Therefore A is Noetherian.